Wizards / Compy Club - ages 4-6

Date: 2026-04-03
Status: in development

Would there be interest in an extra-curricular STEM club for children aged 4-10 in Dubai Harbour / Dubai Marina area? Each supervised session would last around 2 hours with breaks. As it is an experimental program, participation is free (though feedback is much appreciated), but capacity is limited.

Topics covered would include computer programming, math, physics, astronomy in a fairly integrated fashion; what you learn in one subject is immediately put to use in another. No prior experience is assumed, children of different ages learn together, children who understand something ahead of others are encouraged to present it to their peers and answer questions. Developing STEM-related presentation skills (oral presentation,as well as technical/academic writing and drawing) is an important objective.

Principles

Individual progression between exercises

- If someone is having fun with something, don't take it away! No rush.
- If someone is having trouble or slow, give them assitance, time if needed.
- Corollary of the two:
    - Progression between exercises must be individual, team re-sync is only possible after beginning of the day or after the big break.
- Being bored is to be avoided.
    - If bored because they get it fast: provide alternative exercise.
    - If bored because they lost the thread:
        - If possible, detect it before the student gets the feeling that "I failed".
        - Do not press it - divert. The attempt helps, even if failed, but it helps only with added time (a week).
        - Provide an alternative exercise, or game.

Methodology

Structure

Start with repeating

Starting the day with repeating a game from last time is in general desirable, re-establishes the context in the heads.

Theory

Teach via pair-programming

- Hard - how to guarantee that they get partners with whom they like doing it?
    - Sometimes it isn't necessarily the issue that the partner is wrong — some kids may have a good moment when they like partner work, and another moment when they hate it.

- Often pair-programming results in asymetric roles: one does it, the other instructs, thinks, etc.
    - Possibly it's better near the end, when there's a bit more complexity, e.g. putting together control sequences to get out of the labyrinth.

Lesson 1 - The Keyboard

Get to the feeling that I don't need be afraid, it's familiar, I can push it, pull it, fold it, it's fine.

Student achievements

- I dare to open the machine, it's mine. I can close it, open it again.
- I can start the game.
- I can choose which game I want to start.
- I can play the keyboard game.
- I can play a game with the mouse.

Didactic goals

Exercises

Theme: getting familiar with the keyboard.

Notes:

- For older groups, we'll add practice for shift-like combinations (shift, ctrl, caps-lock, delete, esc)
- For everyone: backspace, enter, navigating with arrows
    - For the little ones, backspace is enough, del is not needed.

Exercise: Choose the same the key

Time: 10 min

Description: The screen shows a keyboard. Once the game starts, a key is marked. The player should hit the same key on the keyboard, that removes the mark, and gives a warm sound effect. The process is repeated with other keys.

This takes you through all the key groups (numbers, letters, punctuation, control).

Note: this one does not aim to teach combining characters to words, rather it leads to interesting, so far unfamiliar characters and keys (punctuation, control).

Exercise: Find the key

Time: 15 min

Description: This time the keys do not lit up on the keyboard, but rather they come up one after the other somewhere above of below the keyboard.

Exercise: Hunt the falling objects

Description: Words are falling from the sky. If you type them fast enough, they disappear, and the next word comes.

Theme: getting familiar with the mouse

Exercise: Pop the bubble

Description: Big bubbles, big mouse pointer. A bubble appears, move the pointer there, click, the bubble pops. Repeat.

Maybe: This game is a crossover between getting familiar with the keyboard and with the mouse. Some balloons must be popped with the mouse. We have to be careful, as aiming to a moving target is a higher skill than just using the mouse.

Exercise: Sticker board

Description: Drag stickers to the screen. Shrink, grow, drop. Drop sticks, the thing cannot me manipulated afterwards. There's an eraser. The eraser is the abstraction of a physical eraser (does not remove entire objects).

Exercise: Pong with the mouse

Description: The classic, controlled by a mouse, played against the computer.

Preparation

Start Compy on the devices, and then suspend them with the power-button. When kids start, they push the power-button, and start from the compy console. Slash-commands are used to start the games.

Lesson 2 - The Maze I

Introduction of the Maze, macros.

Experience that I give commands, and the machine obeys.

Student achievements

- I have improved - now I can do something faster, that I did slow the last week.
- I managed to get through the maze - I won the game.
- I could gain superpowers by creating macros - I could reduce my workload, or I could do the same thing faster.
    - Note: for this, we may want to time the thing, or create artificially annoying repetitive tasks
- I could make it fully automated - now I don't need to sit in the driving seat of a however well automated machine, I can just prepare the instructions, and observe the machine doing it fully autonomously
    - The differentiation between the "superpowers via macros", and the "fully automated" can be artificially enhanced by limiting macros to 3 or max 4 steps.

Didactic goals

- Dissociation of control from the time of its effect
- Concept of memory (store and recall control instructions)

Exercises

Currently implemented

Lesson 3 - The Maze II

Continue with the maze, introduce basic editor, full automation.

Experience a higher level of command - I can give it a list of commands and it will do all.

Student achievements

- I have a power to prescribe for a thing what to do, and it does it (I'm the boss).
- I succeeded figuring out a complex sequence of instructions without hitting the wall.

Didactic goals

- First introduction of the editor concept.
- Statement concept (macro definition statement, execution statement)
- Can define higher level behavior.
- Experience building up complex behavior from multiple simpler blocks

Exercises

Exercise: Play the maze physically

Breaking down complexity can be practiced physically.

Maybe enable sticking some cards to a stripe of paper.

This may have to be designed to require just very simple accomplishments from the younger age group.

Exercise: Control a LEGO robot

Play the whole thing with a LEGO robot (1 per group, together), or a physical track

Exercise: Write code to do it on the machine

The teacher establishes the capability of using the editor one-by-one, in person separately.

The system provides multiple labyrinths, with slightly elevating difficulty.

If we have a 5-6 year old, then their exercise is to learn how to explain using the editor.

Lesson 4

Draw with commands - commands are a representation of the drawing.

Student achievements

- I have created something cool, that is new, and that is fully what I created.
- I can bring something into existence by describing it
- I have figured out some puzzles

Didactic goals

- Digital representations; a symbolic description is another form of an object

- A number can represent length
- The power of creation via textual instructions - the first step towards wizardry.

Exercises

Exercise: Play the maze physically, but now draw

Repeat the “Play the maze” exercise, but instead of walking, now do it with a pencil. Someone says an instruction, the others execute it with a pencil.

Exercise: Copy an existing one

The sample drawings are simple abstract figures on paper with only rectangles. The exercise is to implement the scripts that result in the same drawing.

Exercise: Create horizontal, vertical bar and rectangle

Experiment with drawing mostly freely, but then asking what does it take to draw something that’s long horizontally, long vertically, or roughly the same size horizontally and vertically.

This exercise presents the opportunity to explain how they figured out that drawing a long horizontal bar requires larger number of east/west steps, for example? The teacher facilitates a conversation that is a resoning about the connection between the shape and the text.

Exercise: Here’s the code, what will it draw?

Some of the code samples are without subroutines, and then some utilize subroutines. There are instances where the ones with subroutines result in the same drawing, just with simpler code.

Lesson 5

Measure and create graph.

Student achievements

- I could play with a physical stopwatch.
- I succeeded physically using the stopwatch, because I got nice straight lines (at least in two instances) on the paper.
- I could predict what will happen.

Didactic goals

- Establish the concept of a time-position diagram.
- Use the time-position diagram to make predictions using a ruler
- Observe the shape of the time-position diagram to reflect “normal” and “not normal” behavior (uniform versus accelerating).

Exercises

Exercise: Estimate length of objects, then try the guess

The computer show some very simple objects (e.g. a long, horizontal rectangle), but without a grid, so you can’t count how long it is. Your taks is to write the code that draws it, hit enter, and the computer will draw it in a different color — if matches, you win.

Exercise: Measure time it takes for an object to hit some milestones

The computer shows 4 horizontal lines, small cars (or just dots) are standing on the left. There's a start, and a reset button. When you click start, then the thing counts 3, 2, 1,… and everything starts moving. You can measure with a stopwatch the time it takes something to get to a milestone. You have a physical paper with an x-y coordinate system, the milestones are pre-marked on the x axis. Your job is to mark your measurements (with the same color) on the y axis. Then mark the point on the graph matching the milestone and the time, and see that there's two colors where there's a straight line, one color where it's not a straight line — and coincidentally, the car for that color is very fast at the end.

Exercise: Use the established graphs to read when the car will arrive at a certain milestone

The program goes the next phase, where it draws another milestone, one that wasn't drawn before — something simple, exactly halfway between another two. Question: when will the car be there. Students make a guess by reading it from their paper diagrams. They can use a ruler to make their estimations by fitting straight lines (to those graphs which are straight.) Then they can reset, run, and measure to verify.

Exercise: Use the established graphs to read where the car will be at a given time

The system asks the student to mark the milestone on the screen (roughly) where the car will be at a given time. Then reset, run. This time the computer shows time during run, and stops after the given time.

Lesson 6

Two compies drive each other, one presents colors on the screen for a time t, the other drives a lego robot (or the robot itself) for that time to a direction prescribed by the color.

Sudent achievements

Didactic goals

- concept of time
- returning results from subroutines
- const variable, reuse of returned value

Exercises

Exercise: Enact the bee's dance

This is a further developed variant of the "play the maze physically". This time there's a scout, who needs to find an object on a physical maze, and come back with a program. Then the group goes, executes the program, and sees if the thing was found. The didactic goal is to demonstrate that this method can be used to represent, store and communicate algorithms.

Lesson 7

Same as lesson 6, but event driven API - the change of color generates an event, which is then used to make 1 discrete step.

Sudent achievements

Didactic goals

- event handling

Exercises

Lesson 8

Build a scanner: compy drives a robot that goes through a labyrinth and displayes the colors of the cells on the screen.

Sudent achievements

Didactic goals

Integrating all of the above.

Exercises

Appendix

Much of the appendix is random notes.

What sciences could also be taught with it?

- could we teach astronomy, physics, etc
- Fetter Gábor - mobile library with simulating roller balls
    - billiard, pinball, labyrinth with accelerometer
    - he is happy to show sources
- epistemology (Ismeretelmélet): what's the difference between model and reality
    - what we think about is always a model
    - models have limits, which we either know or not
    - when we speak about something being true or not, is not an absolute thing, but is valid only in the context of a model
    - it is worthwhile to study physics through simulations — which is studying models
    - **there are useful false beliefs**
        - it's not a problem that the arial photo is different from the metro map - you just accept that one model has issues, but is enough for what you need
    - science is not about finding the truth - it's about finding out what the limitations of the model
        - when I'm building a house, flat-earth is a decent model
        - when I'm navigating a ship, less so
    - doing it well: exercise the model, and see that it behaves similarly to reality
        - e.g. implement the dropping of a ball in a computer simulation, and observe that it's similar to reality
            - have a hypothesis: this is how gravitation works
            - write a simulation

            - put simulation and real experiment next to each other, see they're similar
            - conclusion: our model works for this setup
    - **doing it bad: doing experiments in the model as if it was reality**
        - **e.g. implement the dropping of a ball in a computer simulation, we look at it in a simulation, and yes, it confirms**
            - **we have the simulation**
            - **we start experimenting in it, and build physics based on the observations**
    - **learning how to do measurement on simulation is a cheap, useful, practical method**
        - **but lieing that it's learning about the world is harmful**
    - **TODO: find Feynman's critique about teaching physics in Brazil -** https://v.cx/2010/04/feynman-brazil-education
        - **give kids data, ask them to synthesize laws**
        - **Feynman showed that these were simulated data**
    - e.g.: use Kepler's equations to do a simulation - then go out see the moon

Further exercise ideas

- Something when data and code is mixed (ants, not bees)
- Something about physics and astronomy

Theme: Keyboard and mouse handling

Exercise: Car-race via typing

Exercise: Pong 2D and Pong 3D

The classic pong game, driven via mouse. Helps developing motor coordination about mouse. Two players, one with the mouse, one with keyboard. Initially the one with keyboard has advantage, later the one with mouse.

Exercise: Prepare the way

An object on a matrix. It starts flashing where it will move (L, R, U, D). You have to click on the cell towards that direction in time, else it crashes.

Exercise: Play with stickers

Like a drawing program, but you can drag ready-made stickers, shrink and grow, and put down.

Exercise 1.4: TODO, something which is more of a math question, and can be done as a group exercise

Exercise 1.5: Some pixel-drawing thing, that is not programming, but the goal is to create stuff

code can make decisions

the program is a communicable thing - try to get someone to draw something by telling them a sequence of steps

Maze

Iterations of the maze

- Simply N, S, E, W
  - Should there be variants?
    - Empty, Small, medium, large?
  - Should there be some variant, that is specifically designed to contain repeated patterns, to sneak in the desire to automate?
- Macros via drawing key images
  - Predefined macros
  - User-defined macros, the good old way (type: macro button, associated hot key, macro sequence, macro button)
- Writing code; the concept: I can write instructions on a piece of paper, the computer can later take it, and do it
  - Write a sequence of characters, very little difference from macros, but introduces an editor feeling, and text drives action feeling
    - Need to teach basic editor functionality
    - Maybe this is a specialized editor with limited feature set, to allow for teaching the proper editor next week
  - Introduce syntax for defining a macro

Notes

- The two different kinds of control may be a good to leave in the final product, to create an opportunity for students to have opinions, discuss, form camps, etc. about which is the *only one* that's good :)
- Idea for the break:
  - when they figure that N, S, E, W is better, get out to the playground, and give the instructions in real life, and get surprised
    - possibly do the same blindfolded with both kinds of instructions
- Bringing in a lego robot, and enable them driving it the same way

Currently implemented

Maze

This game is for teaching small children some basic scripting.

The player's task is to guide a turtle through a maze using sequences of

(single-letter) instructions. These can be either absolute

N - North

S - South

W - West

E - East

or relative

L - left

R - right

F - forward

B - backward

The maze is defined in a separate lua file as an array of strings, with the # sign denoting walls. The initial position and orientation of the turtle is denoted by an absolute direction letter in the corresponding place (see above). There can be only one of these. Destinations are denoted by the * sign. There can be more than one. The map is centered in the screen.

For graphics, please use the turtle from the “turtle” example project. Destinations are denoted by a filled red circle big enough to fit the turtle inside. The background of the game is non-bright white, i. e. Color[Color.white], wall squares are bright blue: Color[Color.blue + Color.bright].

1. Remote control

There is no (obvious) scripting here, just real-time remote control: hitting keys immediately adds the corresponding action to the execution queue, which is consumed as soon as all previous actions have been executed.

Each valid command is acknowledged with a “ping” sound. Relative commands are 1 second animations of either turning the turtle 90 degrees in the corresponding direction or moving it forward or backward one square. Absolute commands are “compiled” into relative commands as follows: if it is executable by one of F or B, it is done so. Otherwise, the nose of the turtle is turned in the corresponding direction first and then the turtle moves forward.

While moving forward, the turtle leaves a thick cyan trace (lineWidth about the same as the turtle’s head) between the centers of the corresponding squares. While moving backward, the turtle leaves no trace.

If a move forward or backward would result in the turtle hitting a wall or moving off the map, the animation is moving the turtle just as much

to touch the side of its squre and then turning her red followed by

playing the “lose” sound. After a half-second pause, the game is reset

to the beginning of the level.

Upon reaching a destination, the “win” sound is played and a half-second

animation reducing the radius of the red circle to zero makes it disappear.

In the first iteration, at this point the game quits.

storytelling method

- You need to tell a story
- Then write a script (forgatókönyv)
- Then write the code, and see that it executes in the pre-planned way

maze ideas

It’s possible to navigate like bees (dancing the map), or ants (navigating via leaving some marks)

experiences with a 10yo

- he liked the examples given without explanation
    - some examples had immediate effect, some not
        - ex. for no immediate effect: setColor(“red”)
- listening to explanations were visibly a suffering
    - even though it was also obvious that he made an honest effort
- looks like those books that are good for adhd people may not be the best for others
- just read a book about how to win lego competitions - small number of examples, lots of explanations
- how about 2 books
    1. examples, hands-on, I can type them in
    2. for those who like read and just fantasy, another one (reads book even before unboxing)
- Dani had access to a computer (not at home yet), but already started reading books, even without accessing the computer
    - is it teachable, that simply to read gives joy?
- principle: examples are physically frequent - 2 a page (explanations are later)
- attempted to teach a bit of logic
    - let’s do minesweeper - good game to teach logic - its logic is simple enough, but not trivial
    - liked it
    - more likely that will do it at home
    - the thought that slow reasoning is stronger than intuition didn’t register by him
    - theoretically ADHD people want to know first how will I benefit from it in 10 seconds

            - successful ones hack themselves setting a series of short term goals
        - he made mistakes in both directions
            - he just clicked via intuition
            - he proved something, and yet asked if he could click
- question: how to train the ability to connect reasoning with action
    - train it… …have multiple such games
    - sokoban
- idea: get them used to this through quick, flashy excercises, ones which cannot generate feeling of hopelessness, repeat week by week

observations

There's two kinds of scams on the education market:

- put the bar very low
- extremely early pick those kids who are fanatics
    - those who do it end up feeling that this is good
    - though only the selection was the contribution of the system

For gymnastics, systems do both: they choose a good ones, and then give them very easy exercises.

Question: how to prevent market incentives to push them to either.

Question: in a very asymetrically informed market, how do we make it a recognizable brand value.

Education is extremely asymetrically informed, the teacher knows what they are doing, the kid have just vibes, but it's easy to manipulate, the parent knows little.

Idea: on competitions, where it's possible, send our students, and it should be known that they came to our club

LEGO robots

- https://pinebricks.nl/readshop/
- https://www.makerzoid.com/
- Jimu Robots
- Possible to do physically measuring tool

Thinking

- functions
    - mapping
    - **returning results from subroutines**
- function composition
- event handling
- the concept of time
- state machine
- **(const) variable (to store values, and use them later)**